

Project 3: Low-Power Tamagotchi
ECE 2564 – Embedded Systems
Virginia Polytechnic Institute of Technology
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GitHub Repository: <https://github.com/ECE2564-S23/proj3-hw12-benjivt>

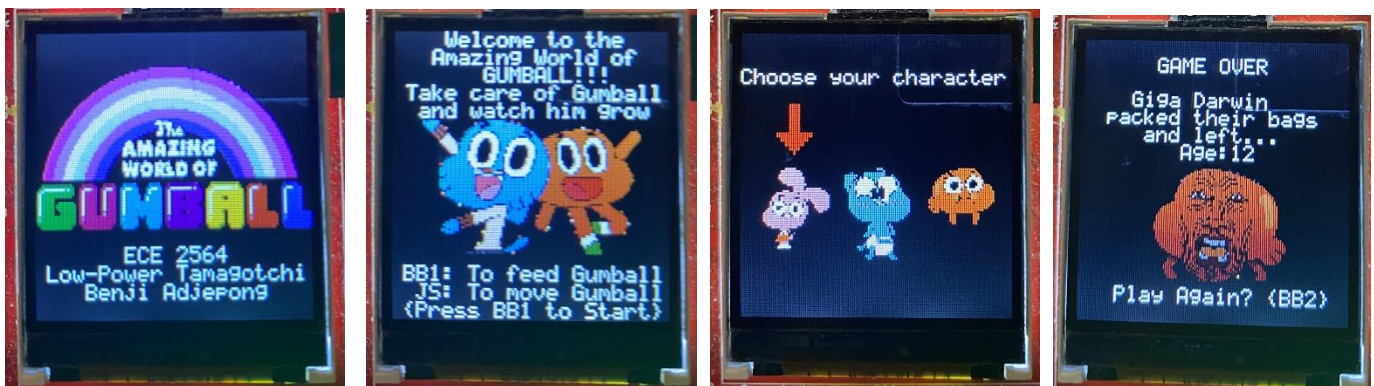
Report Summary

The objective of this report is to present the details of the Low-Power Tamagotchi project. A Tamagotchi is a virtual pet game where a player must take care of the pet to keep it alive. This game was developed by implementing finite state machines and pointers to application struct variables.

Project Description

The Low-Tamagotchi project features a virtual pet game, where a user must feed and play with their pet and lose if they neglect their pet for long. This game is themed after the cartoon *The Amazing World of Gumball* and as such the playable pets are Gumball, Darwin, and Anais. Before starting the actual game, there is a select screen to pick between these characters. After picking a character, the user must feed and play with the character to keep the game going. Pressing Boosterpack button 1 feeds the pet and gives energy while tapping left or right on the joystick moves the pet left or right. Moving the pet increases happiness but decreases energy over time, and the pet cannot move without energy. Every three seconds the pet's age increments but both energy and happiness are decremented, prompting the user to play with their pet. Throughout the game, the pet can grow to different life stages based on its age, energy, and happiness. The game ends once both energy and happiness have reached zero. Below are all the different screens used in this project.

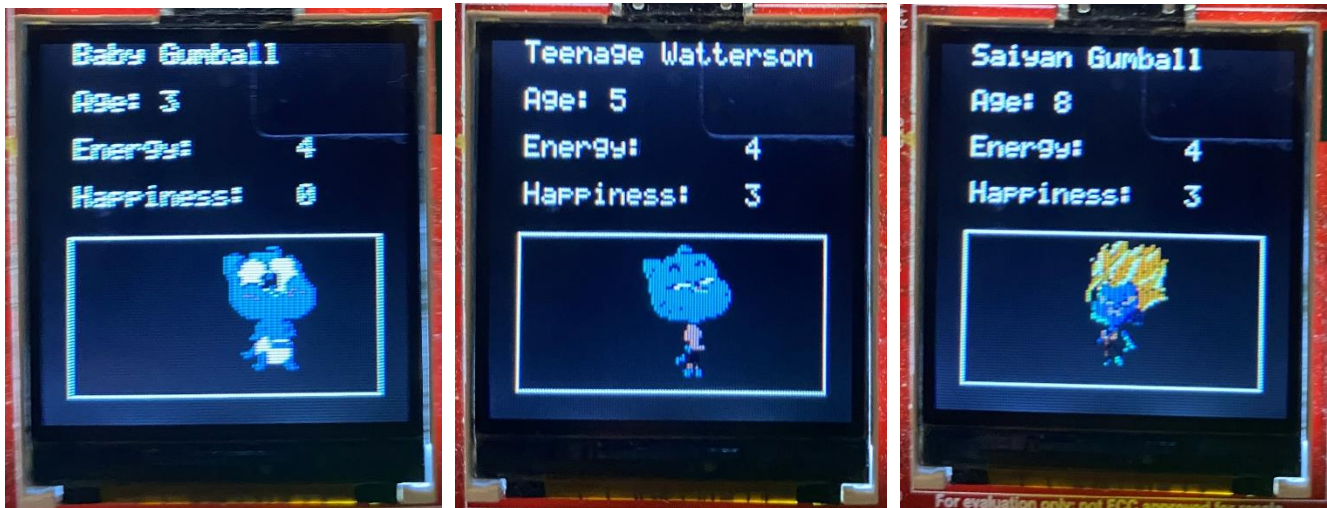
- Title Screen, Instruction Screen, Preset Screen Game Over Screen



- Anais Gameplay



- Gumball Gameplay



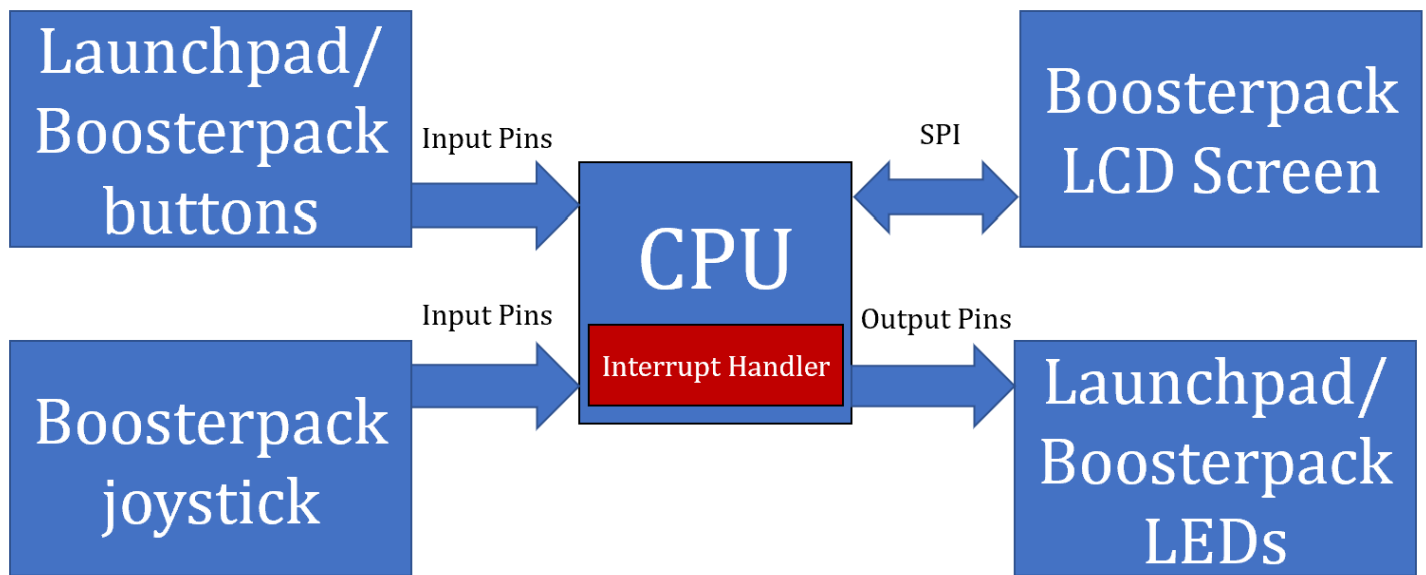
- Darwin Gameplay



System Architecture

The Low-Power Tamagotchi project operates on an interrupt-based architecture, meaning the CPU is always in low-power/sleep until an interrupt is asserted, waking up the CPU to perform a task. The system architecture consists of SPI communication protocol and GPIO connections for data transmitting/receiving. However, these comm protocols are interrupted based and send an interrupt to the CPU whenever an input is registered. The GPIO port and pins are used to receive inputs from peripherals such as the Launchpad buttons and the Boosterpack buttons and joystick. Likewise, the GPIO ports and pins are also used to send outputs to peripherals such as Launchpad and Boosterpack LEDs. The SPI comm protocol is serial and full duplex meaning that data can be transmitted and received in two directions simultaneously. The SPI protocol established a connection with the Boosterpack LCD screen and enabled graphics to be displayed on it.

System Architecture Diagram



Code Quality

It is expected that the project was developed in high quality code since well-written code can be easily tested, updated, and shared. For this project, the quality of the code is evaluated as shown below.

	Description	Expected Score
Comments	Code is reasonably commented so that any person can understand the code	10/10
No Global Variables	No global variables are used, except for custom images using Image Reformer	20/20
No Numeric Values	Macros and constants are used instead of hard-coded numbers, except for UI/Graphics elements	20/20
No Long Functions	No function exceeds 50 lines of code, within reason. Comment lines and empty lines do not contribute to this total.	30/30
Non-blocking code	Code is completely non-blocking. Pressing JSB toggles Green Boosterpack LED at any time.	20/20

Bonus Points

This project had several opportunities to score additional bonus points. The bonus sections that were implemented were custom images for the title screen and game over screen, sprite work, and Tamagotchi presets. The images for these parts were obtained from free PNG websites online. The Tamagotchi presets were implemented using various finite state machines for each Tamagotchi and life stage.

Bonus	Description	Expected Points
Custom Images	Custom images for title and game over screen	10/10
Sprite Work	Use custom images to represent Tamagotchi life cycle	20/20
Presets	Gives the user a choice of different Tamagotchis at the start of the game. Implemented with <i>Sprite Work</i> Bonus	20/20